

```
In [2]: import tensorflow as tf
import matplotlib.pyplot as plt
```

2026-02-22 12:43:26.527058: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

Loading and Visualizing the data

```
In [3]: # Load the MNIST dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data(
# print(x_train[0])
# Normalize
x_train = x_train / 255.0
x_test = x_test / 255.0

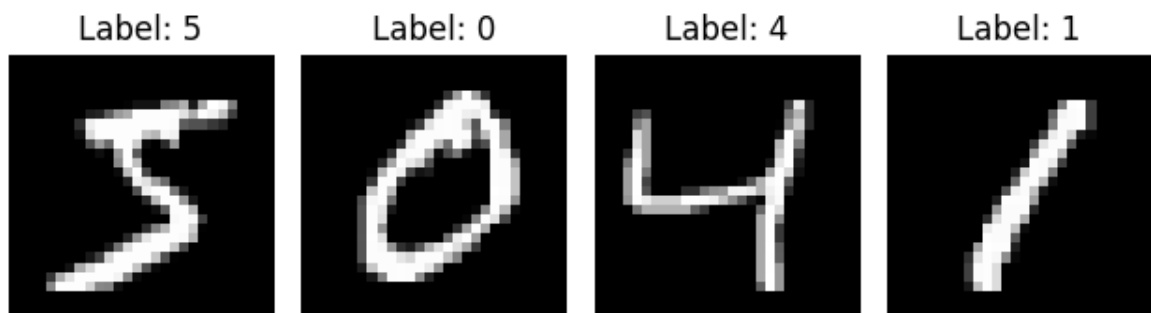
print("Training data shape:", x_train.shape)
print("Testing data shape:", x_test.shape)

for i in range(4):
    plt.subplot(1, 4, i + 1)
    plt.imshow(x_train[i], cmap="gray")
    plt.title(f"Label: {y_train[i]}")
    plt.axis("off")

plt.tight_layout()
plt.show()
```

Training data shape: (60000, 28, 28)

Testing data shape: (10000, 28, 28)



Making the NN and Training it

```
In [4]: model = tf.keras.models.Sequential([
    tf.keras.layers.Flatten(input_shape=(28, 28)),
    tf.keras.layers.Dense(64, activation='relu'),
    tf.keras.layers.Dense(10, activation='softmax')
])

model.compile(loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

```

history = model.fit(x_train, y_train, epochs=5, validation_data=(x_test,
test_loss, test_accuracy = model.evaluate(x_test, y_test, verbose=2)
print(f'Test accuracy: {test_accuracy}')

train_loss = history.history['loss']
train_accuracy = history.history['accuracy']
val_loss = history.history['val_loss']
val_accuracy = history.history['val_accuracy']

# Plot test and training accuracy and loss

plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(train_loss, label='Training Loss')
plt.plot(val_loss, label='Validation Loss')
plt.title('Loss over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.subplot(1, 2, 2)
plt.plot(train_accuracy, label='Training Accuracy')
plt.plot(val_accuracy, label='Validation Accuracy')
plt.title('Accuracy over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.tight_layout()
plt.show()

```

/home/ardo/mlem/Study/Sem8/.venv/lib/python3.12/site-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

super().__init__(**kwargs)

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

I0000 00:00:1771744411.219434 3946 gpu_device.cc:2020] Created device /job:localhost/replica:0/task:0/device:GPU:0 with 2613 MB memory: -> device: 0, name: NVIDIA GeForce GTX 1650, pci bus id: 0000:01:00.0, compute capability: 7.5

Epoch 1/5

2026-02-22 12:43:33.110712: I external/local_xla/xla/service/service.cc:163] XLA service 0x7ff5e40064c0 initialized for platform CUDA (this does not guarantee that XLA will be used). Devices:

2026-02-22 12:43:33.110738: I external/local_xla/xla/service/service.cc:171] StreamExecutor device (0): NVIDIA GeForce GTX 1650, Compute Capability 7.5

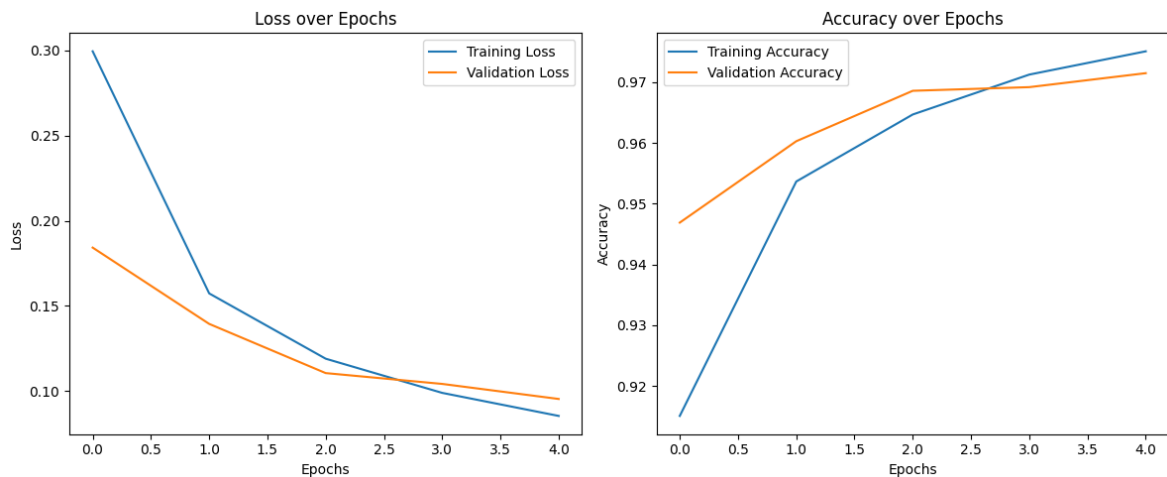
2026-02-22 12:43:33.131126: I tensorflow/compiler/mlir/tensorflow/utils/dump_mlir_util.cc:269] disabling MLIR crash reproducer, set env var `MLIR\_CRASH\_REPRODUCER\_DIRECTORY` to enable.

2026-02-22 12:43:33.237989: I external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:473] Loaded cuDNN version 91900

73/1875 ————— **3s** 2ms/step - accuracy: 0.5344 - loss: 1.5844

I0000 00:00:1771744414.024603 5416 device_compiler.h:196] Compiled cluster using XLA! This line is logged at most once for the lifetime of the process.

1875/1875 ————— **7s** 3ms/step - accuracy: 0.9151 - loss: 0.29
 94 - val_accuracy: 0.9469 - val_loss: 0.1842
 Epoch 2/5
1875/1875 ————— **5s** 2ms/step - accuracy: 0.9537 - loss: 0.15
 74 - val_accuracy: 0.9603 - val_loss: 0.1395
 Epoch 3/5
1875/1875 ————— **5s** 2ms/step - accuracy: 0.9647 - loss: 0.11
 90 - val_accuracy: 0.9686 - val_loss: 0.1106
 Epoch 4/5
1875/1875 ————— **5s** 2ms/step - accuracy: 0.9713 - loss: 0.09
 90 - val_accuracy: 0.9692 - val_loss: 0.1042
 Epoch 5/5
1875/1875 ————— **5s** 2ms/step - accuracy: 0.9751 - loss: 0.08
 55 - val_accuracy: 0.9715 - val_loss: 0.0954
 313/313 - 1s - 2ms/step - accuracy: 0.9715 - loss: 0.0954
 Test accuracy: 0.9714999794960022



Saving the weights and printing them

```
In [5]: model.save_weights("mnist.weights.h5")
```

```
In [6]: import h5py

file = h5py.File("mnist.weights.h5", "r")

for key in file.keys():
    print(key)

def print_structure(name, obj):
    print(name)

file.visititems(print_structure)
```

```
layers
optimizer
vars
layers
layers/dense
layers/dense/vars
layers/dense/vars/0
layers/dense/vars/1
layers/dense_1
layers/dense_1/vars
layers/dense_1/vars/0
layers/dense_1/vars/1
layers/flatten
layers/flatten/vars
optimizer
optimizer/vars
optimizer/vars/0
optimizer/vars/1
optimizer/vars/2
optimizer/vars/3
optimizer/vars/4
optimizer/vars/5
vars
```

In [7]: **import** numpy **as** np

```
kernel1 = np.array(file["layers"]["dense"]["vars"]["0"][:])
bias1 = np.array(file["layers"]["dense"]["vars"]["1"][:])

kernel2 = np.array(file["layers"]["dense_1"]["vars"]["0"][:])
bias2 = np.array(file["layers"]["dense_1"]["vars"]["1"][:])

print(kernel1.shape)
print(bias1.shape)

print(kernel2.shape)
print(bias2.shape)
```

(784, 64)

(64,)

(64, 10)

(10,)

Weights for first hidden layer

In [8]: kernel1[:,0]

```
Out[8]: array([ 0.05785387,  0.06126907, -0.06040148, -0.03314932, -0.02935267,
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```

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0.06071984, -0.02747665, -0.0508942 , -0.02787503, -0.00853734,  
0.01173689, -0.00979199, 0.06127112, -0.05500879, 0.04464243,  
-0.01210211, 0.04322492, 0.0587069 , 0.06235657, -0.02470851,  
0.01202869, 0.01188562, 0.07513458, 0.06557624, -0.00886648,  
0.03387047, -0.03533671, 0.02564 , -0.029486 , 0.01839666,  
0.00802242, 0.01452449, -0.0496982 , -0.05006805, -0.0755081 ,  
-0.08201823, -0.0386759 , 0.0924258 , -0.00942532, -0.07495473,  
0.02597316, -0.04947381, -0.05048798, -0.03661725], dtype=float32
```

2)

Bias for first hidden layer

In [9]: bias1

```
Out[9]: array([ 0.09515617, -0.16854578, 0.03957985, -0.24464414, 0.05953605,  
-0.02980527, 0.07394806, 0.07802063, 0.1404661 , -0.20266764,  
-0.2214548 , -0.14675654, -0.08690672, -0.01978087, -0.20430289,  
-0.18470758, -0.14021622, 0.2739049 , -0.04149286, -0.28260082,  
0.04863944, -0.26252422, -0.01987718, 0.06257852, -0.08743436,  
-0.00833195, 0.02790745, -0.0480802 , 0.04092814, 0.11087266,  
-0.13375452, 0.24066186, 0.21312635, 0.02178879, 0.01021563,  
-0.10557787, -0.26352847, -0.0496706 , 0.00921682, 0.12007748,  
-0.01713181, -0.20537436, 0.07473799, 0.0714574 , -0.01302372,  
-0.11763341, -0.11212406, -0.20765932, 0.05033552, -0.22951198,  
-0.06027036, -0.02396407, -0.13383946, -0.29678857, 0.0465657 ,  
-0.18066223, -0.10719177, 0.02805756, 0.04770682, -0.2824266 ,  
0.01409796, -0.0022136 , -0.05507494, -0.23943992], dtype=float32
```

2)

Using a double hidden layer 16 neuron architecture

```
In [10]: model = tf.keras.models.Sequential([
    tf.keras.layers.Flatten(input_shape=(28, 28)),
    tf.keras.layers.Dense(16, activation='relu'),
    tf.keras.layers.Dense(16, activation='relu'),
    tf.keras.layers.Dense(10, activation='softmax')
])

model.compile(loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))

test_loss, test_accuracy = model.evaluate(x_test, y_test, verbose=2)
print(f'Test accuracy: {test_accuracy}')
```

```
Epoch 1/5
1875/1875  7s 3ms/step - accuracy: 0.8717 - loss: 0.44
14 - val_accuracy: 0.9160 - val_loss: 0.2837
Epoch 2/5
1875/1875  5s 2ms/step - accuracy: 0.9278 - loss: 0.24
99 - val_accuracy: 0.9338 - val_loss: 0.2287
Epoch 3/5
1875/1875  5s 2ms/step - accuracy: 0.9379 - loss: 0.21
76 - val_accuracy: 0.9394 - val_loss: 0.2066
Epoch 4/5
1875/1875  5s 2ms/step - accuracy: 0.9428 - loss: 0.19
92 - val_accuracy: 0.9424 - val_loss: 0.1992
Epoch 5/5
1875/1875  5s 2ms/step - accuracy: 0.9463 - loss: 0.18
52 - val_accuracy: 0.9426 - val_loss: 0.1965
313/313 - 1s - 2ms/step - accuracy: 0.9426 - loss: 0.1965
Test accuracy: 0.9426000118255615
```